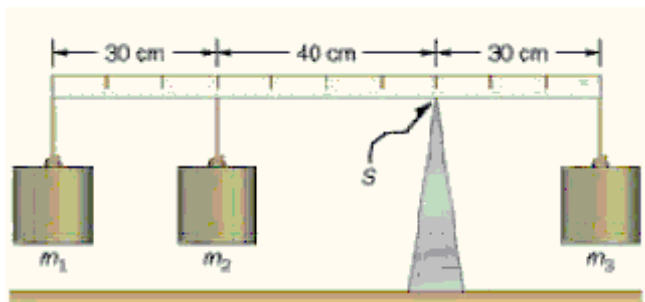
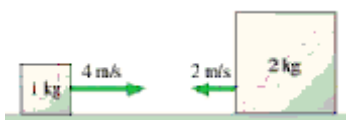


- 1) A disk, a hoop and a sphere are released from the same height on an inclined plane. Which one wins the race. 1) _____
 A) the sphere B) they arrive at the same time
 C) the disk D) the hoop
- 2) Ten seconds after an electric fan is turned on, the fan rotates at 300 rev/min. Its average angular acceleration is (hint: initial angular velocity is 0). 2) _____
 A) 50 rev/s/s B) 3.14 rad/s/s
 C) 30 rad/s/s D) 30 rev/s/s
- 3) An 1100-kg car traveling at 27.0 m/s starts to slow down and comes to a complete stop in 578 m. What is the magnitude of the average braking force acting on the car? 3) _____
 hint: you can use the work-energy theorem.
 A) 550 N B) 690 N C) 410 N D) 340 N
- 4) On its own, a certain tow-truck has a maximum acceleration of 3.0 m/s^2 . What would be the maximum acceleration when this truck was towing a bus of twice its own mass? 4) _____
 hint: new system is the truck + its load (new mass)
 A) 1.5 m/s^2 B) 1.0 m/s^2 C) 2.0 m/s^2 D) 2.5 m/s^2
- 5) A 10,000-kg rocket blasts off from earth with a uniform upward acceleration of 2.00 m/s^2 and feels no air resistance. The upward thrust force its engines must provide during this acceleration is closest to hint: 2 forces are acting on the rocket. Use Newton's second law. 5) _____
 A) 980,000 N. B) 118,000 N. C) 20,000 N. D) 78,000 N.
- 6) $m_3 = 5\text{kg}$, $m_2 = 2\text{kg}$. what is m_1 ? (neglect mass of the stick) 6) _____



- A) 3kg B) 4kg E) 5kg C) 1kg D) 2kg

- 7) The two boxes (1kg , 4m/s @right and 2kg , 2m/s @ left) are sliding along a frictionless surface. They collide and stick together. Afterward, the velocity of the two boxes is 7) _____



- A) 1m/s @ left B) 2m/s @ left
 C) 0 m/s D) 2 m/s @ right E) 1 m/s @ right

8) If electricity costs 25.00¢/kWh (kilowatt-hour) in NYC, how much would it cost you to run a 1,500 Watts unit for 8.0 hours per day for 4.0 weeks? 8) _____
 A) \$150 B) \$12 C) \$84 D) \$25

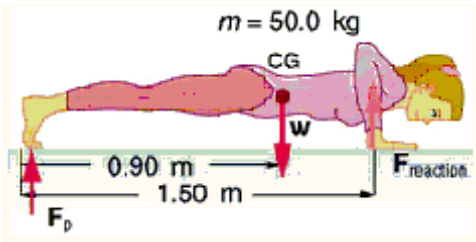
9) Find the net work done by friction on a box that moves in a complete circle of radius 1.82 m on a uniform horizontal floor. The coefficient of kinetic friction between the floor and the box is 0.25, and the box weighs 65.0 N. 9) _____
 A) 1800 J B) 0 J C) 370 J D) 190 J

10) A 2.5-kg stone is released from rest and falls toward Earth. After 4.0 s, the magnitude of its momentum is: 10) _____
 A) 24kgm/s B) 39kgm/s C) 78kgm/s D) 98 kgm/s

11) A tennis ball is tossed straight up with a speed of 20 m/s. What is the height reached? 11) _____
 (g=10)
 A) 1m B) 5m C) 20m D) 10m

12) One revolution per minute is about 12) _____
 A) 0.0524 rad/s B) 0.0524 rad/s
 C) 6.28 rad/s D) 0.105 rad/s

13) What is the force applied by the ground on her hands? She is in equilibrium. g=10ms/s/s 13) _____
 hint:: take the fulcrom (axis rotation) at her feel. Sum(CCW) = Sum(CW)



A) 75N B) 30N C) 450N D) 300N

14) The cart's change of momentum Δp is (the fist figure is before and the second figure is after). 14) _____



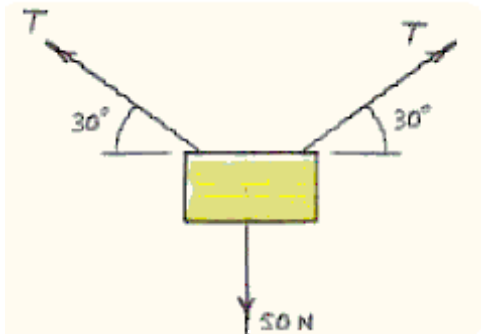
A) - 10kgm/s B) 10 kgm/s
 C) 20kgm/s D) 30 kgm/s
 E) 10kgm/s

15) A hammer and a feather are released from the same height on the Moon. Which one reach the ground first. 15) _____
 A) they arrive at the same time B) the hammer
 C) not enough information D) the feather

16) A mosquito and a truck have a head-on collision. Splat! Which has a larger change of momentum? 16) _____
 A) They have the same change of momentum B) the truck
 C) the mosquito D) can't say. we need the initial speeds

17) The block is in equilibrium. What is the tension T ?

17) _____



- A) 5N E) 10N B) 50N C) 100N D) 25N

18) A car has 1,000,000 joules of kinetic energy and is moving along a straight line. If its speed increases by a factor of 3 (3 times faster) its kinetic energy is (no friction)

18) _____

- A) 9,000,000J B) 2,000,000J C) 3,000,000J D) 6,000,000J

19) A ball is tied to the end of a cable of negligible mass. The ball is spun in a circle with a radius 2.00 m making 7.00 revolutions every 10.0 seconds. What is the magnitude of the acceleration of the ball?

19) _____

- A) 14.8 m/s^2 B) 74.2 m/s^2 C) 29.3 m/s^2 D) 38.7 m/s^2 E) 67.9 m/s^2

20) A child pulls on a wagon with a force of 75 N. If the wagon moves a total of 42 m in 3.1 min, what is the average power delivered by the child?

20) _____

- A) 21 W B) 17 W C) 26 W D) 22 W

Mechanics

$v_x = v_{x0} + a_x t$	a = acceleration
$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$	d = distance
$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$	E = energy
$\vec{d} = \frac{\Sigma \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$	F = force
$ \vec{F}_f \leq \mu \vec{F}_n $	f = frequency
$a_c = \frac{v^2}{r}$	h = height
$\vec{p} = m\vec{v}$	I = rotational inertia
$\Delta \vec{p} = \vec{F} \Delta t$	K = kinetic energy
$K = \frac{1}{2} m v^2$	k = spring constant
$\Delta E = W = F_{\parallel} d = F d \cos \theta$	L = angular momentum
$P = \frac{\Delta E}{\Delta t}$	l = length
$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$	m = mass
$\omega = \omega_0 + \alpha t$	P = power
$x = A \cos(\omega t) = A \cos(2\pi f t)$	p = momentum
$x_{cm} = \frac{\Sigma m_i x_i}{\Sigma m_i}$	r = radius of separation
$\alpha = \frac{\Sigma \tau}{I} = \frac{\tau_{net}}{I}$	T = period
$\tau = r_{\perp} F = r F \sin \theta$	t = time
$L = I \omega$	U = potential energy
$\Delta L = \tau \Delta t$	v = speed
$K = \frac{1}{2} I \omega^2$	W = work done on a system
$ \vec{F}_s = k \vec{x} $	x = position
	α = angular acceleration
	μ = coefficient of friction
	ϕ = angle
	τ = torque
	ω = angular speed
	$U_s = \frac{1}{2} k x^2$
	$\Delta U_g = m g \Delta y$
	$T = \frac{2\pi}{\omega} = \frac{1}{f}$
	$T_s = 2\pi \sqrt{\frac{m}{k}}$
	$T_p = 2\pi \sqrt{\frac{I}{g}}$
	$ \vec{F}_g = G \frac{m_1 m_2}{r^2}$
	$\vec{g} = \frac{\vec{F}_g}{m}$
	$U_g = -G \frac{m_1 m_2}{r}$

21)

Answer Key

Testname: FINAL_24

- 1) A
- 2) B
- 3) B
- 4) B
- 5) B
- 6) C
- 7) C
- 8) C
- 9) D
- 10) D
- 11) C
- 12) D
- 13) D
- 14) D
- 15) A
- 16) A
- 17) B
- 18) A
- 19) D
- 20) B
- 21)